Reg. No. | | | | | | | | | | | [18EC51]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

PRINCIPLES OF ELECTROMAGNETICS

Time:3 Hours **Maximum Marks:70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Derive the expression for Electric field due to infinite line charge	**07**
	b.	Let the charge Q1 = 25 nC be located at A(4, -2, 7) and charge Q2 = 60 nC be at B(-3,4,-2). Find $\vec{E}$ at C(1, 2, 3)	**07**
2.	**a.**	State and prove uniqueness theorem.	**07**
	b.	Two parallel conducting discs are separated by distance 5 mm at z = 0 and z = 5mm. If V = 0 at z = 0 and V = 100V at z = 5 mm, Find the charge densities on the discs.	**07**
3.	**a.**	State and prove Ampere's circuital law	**07**
	b.	Evaluate both sides of the Stoke's theorem for the field $\vec{H} = 6xy\vec{a_x} - 3y^2\vec{a_y}$ A/m and the rectangular path around the region, $2 \le x \le 5, -1 \le y \le 1, z = 0$. Let the positive direction of $\vec{dS}$ be $\vec{a_z}$	**07**
		OR	
4.	**a.**	Discuss the boundary conditions at the interface between two media of different permeabilities.	**07**
	b.	Derive en expression for the force on a differential current element placed in a magnetic field.	**07**
5.	**a.**	List Maxwell's equations for time varying fields in differential and integral forms.	**07**
	b.	In a given lossy dielectric medium, conduction current density J_c = 0.02 sin 109t (A/m^2). Find the displacement current density if σ = 103 S/m and ε_r = 6.5.	**07**
		OR	
6.	**a.**	State and prove Poynting theorem	**07**
	b.	Wet marshi soil is characterized by $\sigma = 10^{-2}$ S/m, ε_r =15 and μ_r = 1. At frequencies 60Hz, 1MHz, 100MHz and 10GHz, Indicate whether soil be considered a conductor, or a dielectric.	**07**

7.	**a.**	Discuss the behavior of Reflection of Plane waves at the surface of a perfect conductor at Normal Incidence.	**07**
	b.	Derive expressions for Transmission coefficient and Reflection coefficient.	**07**
		OR	
8.	**a.**	Write a short note on Standing Wave Ratio	**07**
	b.	Travelling $\vec{E}$ and $\vec{H}$ waves in free space (region 1) are normally incident on the interface with a perfect dielectric (region 2) which εr =3. Compare the magnitudes of the incident, reflected and Transmitted $\vec{E}$ and $\vec{H}$ at the interface	**07**